Xijia "Polina" Zhang

zhang.xijia@gatech.edu Zhang Xi-Jia xijia.me Xijia Polina Zhang polinazhang

Education

Georgia Institute of Technology Aug. 2029
Ph.D. in Robotics

University of Michigan May. 2024
B.S.E in Computer Science

Publications

"Model-Agnostic Policy Explanations with Large Language Models"
Zhang Xi-Jia, Yue Guo, Shufei Chen, Simon Stepputtis, Matthew Gombolay, Katia Sycara, Joseph Campbell COLM 2025 (**top 5%**)

– Propose a method for generating natural language explanations of agent behavior based only on observed states and actions, without access to the agent's underlying model.

"Towards Human-Free Semantic Interpretability in Reinforcement Learning via Vision-Language Models" *(in submission)*
Zhaoxin Li*, **Zhang Xi-Jia***, Batuhan Altundas, Letian Chen, Rohan Paleja, Matthew Gombolay

– Develop an automated RL framework that leverages foundational models for semantic feature extraction and interpretable tree-based models for policy optimization.

"Learning Effective Action Advising in the Face of Changing Rewards"
Yue Guo, **Zhang Xi-Jia**, Simon Stepputtis, Joseph Campbell, Katia Sycara CoLLAs 2024 (**Oral**)

– Enable the teacher policy to continually learn and adapt its reward function through ongoing observation of the student when providing action advices.

"Teaching the Teacher: Enhancing Human-to-Robot Skill Demonstration with Live Foundation Model and Augmented Reality Feedback"
Nina Moorman, Matthew Luebbers, **Zhang Xi-Jia**, Zulfiqar Zaidi, Marcus Lau, Yixing Yao, Megan Langwasser, Letian Chen, Sanne van Waveren, Matthew Gombolay HRI 2026

– Support non-expert users in providing kinesthetic demonstrations through language explanations and policy visualizations.

"Communication and Verification in LLM Agents towards Collaboration under Information Asymmetry
Run Peng, Ziqiao Ma, Amy Pang, Sikai Li, **Zhang Xi-Jia**, Yingzhuo Yu, Cristian-Paul Bara, Joyce Chai COLM 2026

– Investigate the performance of LLM agents in human-robot collaboration tasks under information asymmetry.

Skills

`Toolchain`	ROS, ROS2, Linux, Docker, Huggingface, RLlib
`Frameworks & Libraries`	Gym, NLTK, Transformers, Pytorch, Scikit-Learn
`Foundation Model`	Fine-tuning, Prompting, Deploying; LoRA, PPO, GRPO
`Robotics Development`	TIAGo, Jaco Arm, Khepera, Fetch, OpenArm; RViz, Gazebo